

Creative Systems Design

CSD2a

Vandaag

- Introductie
- Eindopdracht
 - Bespreken
 - Voorbeelden

Over CSD

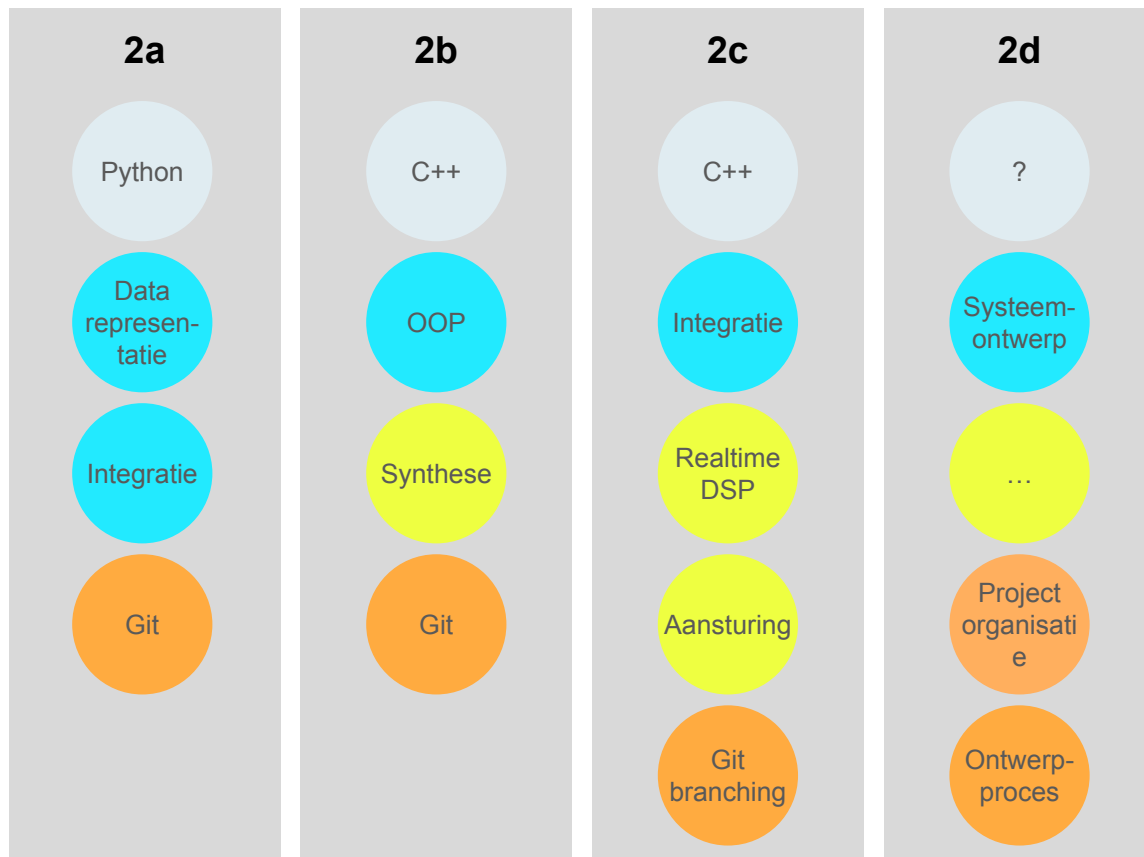
- Creative coding in verschillende MT contexten
- Techniek is dienend (*maar je moet je technieken natuurlijk wel beheersen*)
- Systeemontwerp is afhankelijk van het doel, context, beschikbare tijd etc.
- Technische ontwikkelingen → blijven ontwikkelen

CSD jaar 2

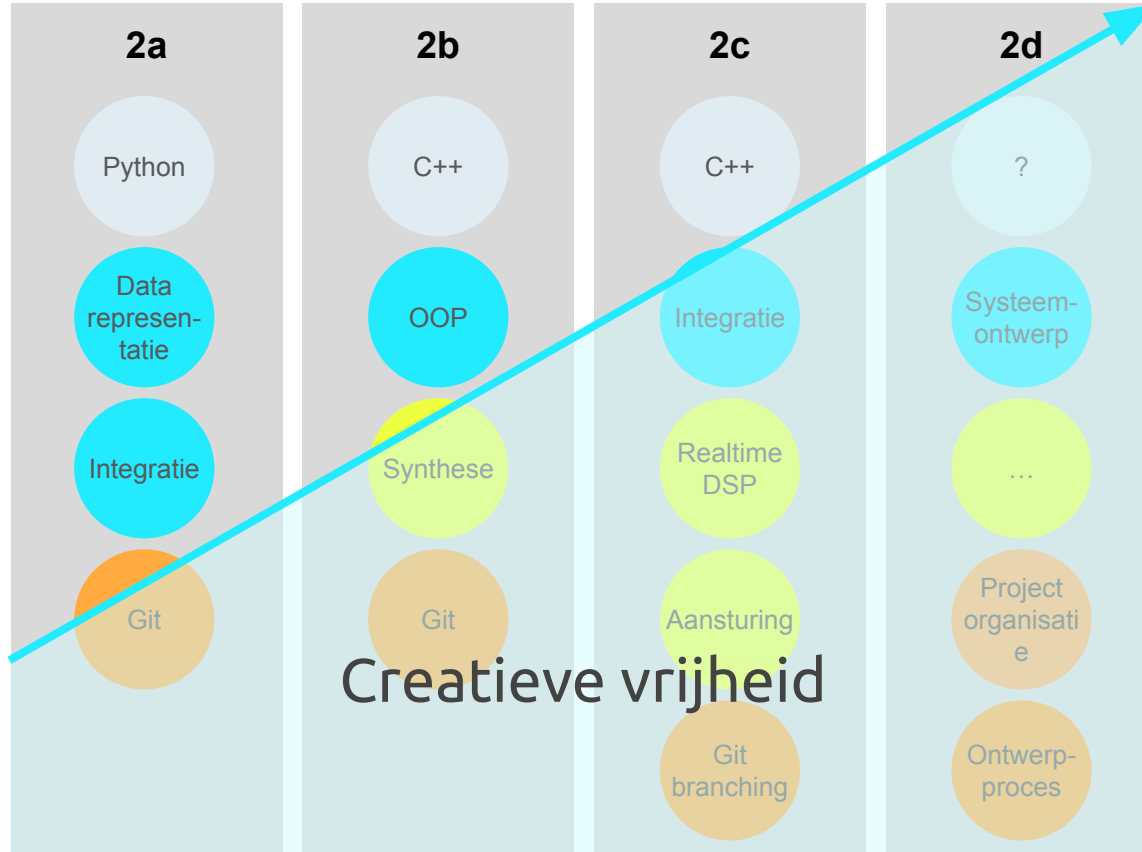
Een aantal kernwoorden:

- Artisticiteit
- Systeemontwerp, programmeertalen en hardware
- Ontwerpproces
- Verslaglegging / documentatie
- Reflectie
- Verantwoording ontwerpkeuzes

Opzet Blok 2a - CSD jaar 2



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CSD 2a - Inzet/investering

- 6 contacturen per week
 - [Systemen bouwen met Python](#) én practicum -> veel ontwerpen en programmeren
 - [CSD Theorie](#) -> de basis
 - [Wiskunde](#) -> de basis van de basis
- 11.5 uren zelfstudie

(5 ECTS voor 8 weken → 17.5 uur per week)

Locatie content

https://github.com/ciskavriezenga/CSD_26-27 in de folder 'csd2a' :

- in **assignments/**
 - de eindopdracht & beoordelingscriteria
 - per week de opdrachten voor zelfstudie (assignments/session[X])
In het vooruit werken is mogelijk, maar de wekelijkse opdrachten kunnen nog worden aangepast
- in **slides/** de pdfs van de tijdens de sessies getoonde slides

Locatie content

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- in **assignments/**
 - de eindopdracht & beoordelingscriteria
 - per week de opdrachten voor zelfstudie (assignments/session[X].md)
In het vooruit werken is mogelijk, maar de wekelijkse opdrachten kunnen nog worden aangepast
 - handouts/ pdfs waaraan gerefereerd wordt
- in **slides/** de pdfs van de tijdens de sessies getoonde slides

en gezien AI met één prompt de eindopdracht genereert

→ alle code al '*gratis en voor niets*':

- in **24-25/** en **25-26/** hele opzet inclusief code voorbeelden van vorige jaren

Vandaag

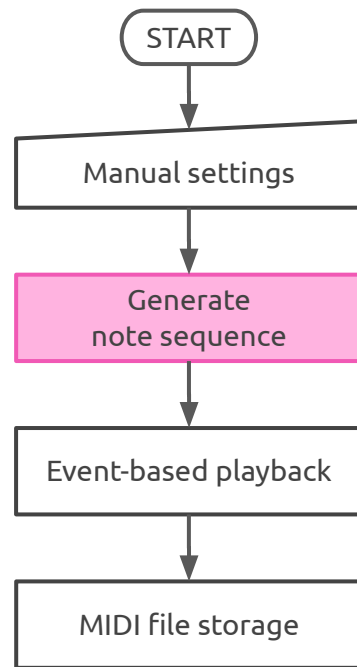
- Introductie
- **Eindopdracht**
 - Bespreken
 - Voorbeelden

CSD2a

Irregular beat generator

A command line application that generates irregular beats according to a custom generation strategy.

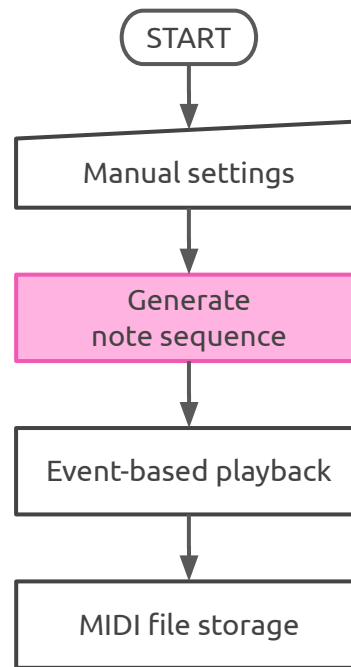
Irregular beat generator



CSD2a

- Week 1
 - **Generation strategy**
 - Flow diagrams / steps
 - Pseudo code

Irregular beat generator

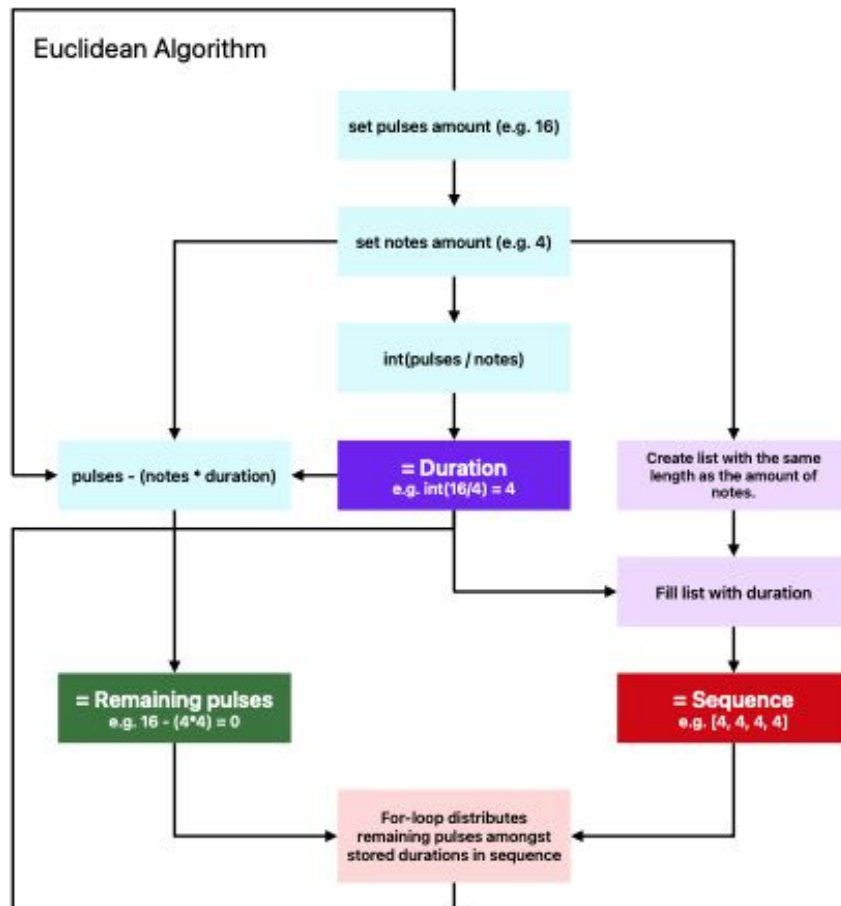


CSD2a

- Week 1

- Generation strategy
- **Flow diagrams / steps**
- Pseudo code

Irregular Beat Generator flow diagram



CSD2a

- Week 1
 - Generation strategy
 - Flow diagrams / steps
 - **Pseudo code**

```
for (let i = 0; i < TotalTime; i++) {  
  let notes = [];  
  notes.append = random option from options;  
  //...
```

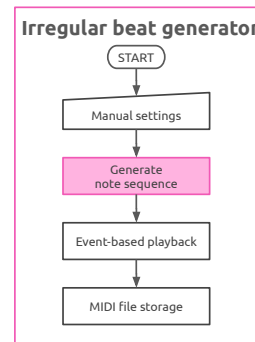
CSD2a

- Week 1
 - Generation strategy
 - Flow diagrams / steps
 - Pseudo code
- Week 2
 - Finalize rhythm generation design based on feedback
 - Playtest the pseudo code

CSD2a

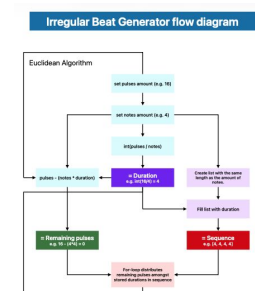
Session 1 & 2

- Example of rhythm generation strategy
- Start main assignment
- Student presentations rhythm generation design



Session 3 - 5

- Student project progress
- Code examples and assignments
 - Building blocks directly added to project



Session 6

Final presentations

```
for (let i = 0; i < TotalTime; i++) {  
    let notes = [];  
    notes.append = random option from options;  
    //...
```


CSD2a

Session 1 & 2

- Example of rhythm generation strategy
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Session 3 - 5

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Session 6

Final presentations

CSD2a

Session 1 & 2

- Example of rhythm generation strategy
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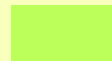
Session 3 - 5

- Student project progress
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Session 6

Final presentations

Overzicht



→ **beoordeling**

1	di.31-08	Introduction & example of rhythm generation strategy	• python up and running, hello *Bleep • design a rhythm generation strategy
2	ma.07-09	Presentations rhythm generation strategies & feedback + next steps / directions	• Github repo & git up and running • Finalize rhythm generation strategy, playtest it pseudo code • python: bleep in a loop & create and play a hard coded rhythm
3	ma.14-09	Hand-in final design & playback with durations vs timestamps	• implement your rhythm generation strategy to generate a note sequence (one 'stem' only for now) and then transform rhythm to timestamps
4	ma.21-09	allow to generate and playback multiple stems by using dictionary to store events	• transform your one-stem rhythm generation to multiple stems rhythm generation
5	ma.28-09	UI & trouble shooting - recap up to now, presentation expectations	• add UI to your basic irregular beat generator (IBG)
6	ma.5-10	Present project up to now , playtest session	Beautify your IBG to the max!
7	ma.12-10	Presentations main assignment	
8	ma.19-10	Vorbereiding blok2b - start C++	• C++ up and running, hello *Sine

Overzicht

→ **beoordeling**

1	di.02-09	Introduction & example of rhythm generation strategy	Github repo, git & python up and running, DIY: design a rhythm generation strategy
2	ma.08-09	Presentations rhythm generation strategies & feedback + next steps / directions	Finalize rhythm generation strategy pseudo code and playtest it (see more elaborate description in opdrachten.md), python: bleep in a loop & create and play a mock rhythm
3	ma.15-09	Hand-in final design & playback with durations vs timestamps	implement your rhythm generation strategy to generate a note sequence from a stem, add an irregular beat generator to your rhythm to timestamps
4	ma.22-09	allow to generate and playback multiple stems by using dictionary to store events	transform your one-stem rhythm generation to multiple stems rhythm generation
5	ma.29-09	UI & trouble shooting loop to v	add UI to your basic irregular beat generator (IBG)
6	ma.06-10	Fancy extra stuff :-D	Beautify your IBG
7	ma.13-10	Present project up to now & presentation expectations & playtest session	Beautify your IBG to the max!
8	ma.20-10	Presentations main assignment	Practicum after final presentation? --> no practicum in that case

Dikke disclaimer ... iets dat niet werkt en/of onleesbaar is kan niet beoordeeld worden.

```
if(not_working || unreadable_code):
    result = NA
```